Q.31 State reasons why the load in a consumer's installation is divided into sub circuits?

Q.32 Discuss the methods of service connection line to a double storyed building?

Q.33 Discuss various types of line supports.

Q.34 Draw the layout of an Indoor Type Substations.

Q.35 Enlist and describe the equipments required for an outdoor Substations.

## SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 A 400 KVA, 22kv/415 V foundation mounted substation is required to supply electrical energy to the one portion of town. Estimate the list of material required for erection of this susbtation.

Q.37 Design a 33kv feeder using 11 m long RCC Poles and calculate the cost of material if the feeder is erected overhead on the basis of existence of 33 kv ovehead line, if the length of proposed feeder is 5km upto a substation where the feeder is terminated

Q.38 Estimate the quantity of the material and cost for giving service connection by using suitable copper conductor to a college having load of 50 KW (Light and fans). Assume distance from 10 m. P.C.C pole of college 40 meters and from support to meter board 18 meters.



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Roll No. ..................

**3rd Sem / Electrical, Power Station Engg., Elect. & Eltx. Engg.**

**Subject:- Estimating & Costing in Electrical Engineering**

Time : 3Hrs. M.M. : 100

## SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 To lower the resistance of ground rod, the chemicals used are

a) Copper sulphate b) Sodium sulphate

c) Nitric Acid d) Sulphuric Acid

Q.2 Voltage rating of a pole mounted substations is

a) 5 KV b) 11 KV

c) 33 KV d) 66KV

Q.3 The melting point of the fuse wire should be________

a) Moderate b) Low

c) High d) Infinite

Q.4 ________ Phase service line is required for Industries

a) 1-$\phi$ b) 3-$\phi$

c) Both (a) and (b) d) None of the above

Q.5 What is the maximum number of oultets, which can be connected in a power sub-circuit?

a) Four b) Three

c) Eight d) Two

Q.6 From safety point of view, the most suitable wiring system is

a) Cleat wiring

b) Casing-Capping wiring

c) Conduit wiring

d) Metal Sheathed wiring

Q.7 The load on each light-fan sub circuit shall be restricted to

a) 800 W b) 1000 W

c) 1200 W d) 1500 W

Q.8 For Large Substations, which type Earth electrodes are recommended

a) Rod b) G.I. Pipe

c) Copper Plate d) Strip

Q.9 The size of LT cable depends on the rating of

a) Feeder b) Transformer

c) Distributor d) Busbar

Q.10 The position of circuit breaker where incoming cable enters the building is determined by

a) Phase b) Neutral

c) Earth d) All of the above

## SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

Q.11 What is Tender?

Q.12 Define Comparative statement.



Q.13 Explain Earnest Money.

Q.14 Expand ELCB

Q.15 Write the full-form of RCCB

Q.16 Define contingency.

Q.17 Name the types of substation

Q.18 What is service Line Connection?

Q.19 What is earthing?

Q.20 Define Estimating and Costing.

## SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 What do you mean by requisition? Make a specimen of requisition form.

Q.22 Discuss the procedure of tender opening.

Q.23 Make a performa for making an estimate.

Q.24 Write a short note on the purpose of Estimating and costing?

Q.25 Name the various accessories used in house wiring.

Q.26 Discuss the casing and capping type of wiring and its advantages and disadvantages

Q.27 What are the factors influencing Earth Resistance.

Q.28 Prepare the list of materials required for earthing. Estimate the cost of material used in Earthing.

Q.29 What is the difference between Light and Power and motor wiring?

Q.30 Draw the wiring diagram to Energy meter, Main switch and Main distribution Board.